CSS183-2 SA01 - Process in Balancing Dataset

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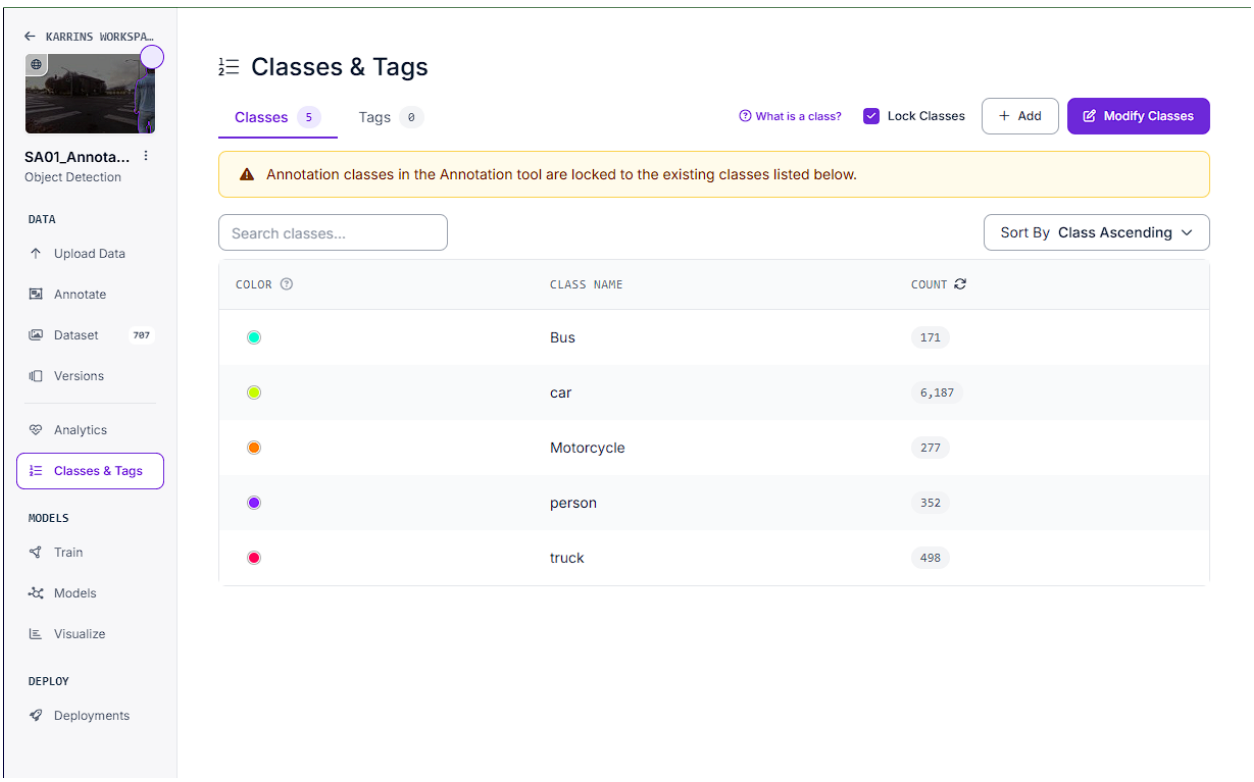
I. Introduction

This paper explains how class imbalance was identified in the dataset and describes the methods used to manage and reduce its impact. In object detection tasks, balanced data is important because machine learning models may become biased toward classes that appear more frequently.



To create these dataset, Highway CCTV videos were first collected from publicly available sources such as Kaggel or Youtube. Frames were then extracted form those videos at a rate of 1 frame per second (1 FPS) to reduce redundancy. After extraction, the frames were uploaded to Roboflow and annotated using the auto-labeling feature (SAM 3 model). Each image was manually reviewed to verify the annotations. Incorrect detections

were adjusted or removed, and low-quality or blurry frames were excluded. After this review process, approximately 700 verified images were retained as the final annotated dataset before augmentation.

Since traffic scenes naturally contain more cars than other vehicles, some level of imbalance is expected. However, it is still necessary to analyze the class distribution and assess whether the imbalance may affect model performance.

II. Identifying Data Imbalance

After the annotation review process, approximately 700 verified images were included in the final dataset before augmentation. To analyze potential imbalance, the class distribution was examined using the Roboflow "Classes & Tags" analytics panel.

The total number of object instances detected across the 700 reviewed images is shown in the table below

CLASS	NUMBER OF INSTANCES
Car	6,187
Truck	498
Person	352
Motorcycle	277
Bus	171

From the results formt the table above, it can be observed that the Car class significantly dominates the dataset. Cars appear 6,187 times, which is substantially higher compared to other classes such as Bus (171 instances) and Motorcycle (277 instances).

This indicates a clear class imbalance. The dataset reflects real-world highway and traffic conditions where cars naturally appear more frequently than other vehicle types.

However, these large difference in instance counts may influence model learning behavior if not properly addressed during training.

III. Reasons for Class Imbalance

There are alot of reasons why there is such a clear class inabalsne in thesee types of datasets. The class imbalance in this dataset is primarily due to the natural distribution of objects in real-world traffic scenes. In urban intersections, cars are significantly more common compared to buses, motorcycles, or trucks. As a result, during frame extraction and annotation, cars appeared more frequently across the collected footage.

Another contributing factor is the video source itself. The dataset was created from publicly available traffic recordings, where typical daily traffic conditions were captured. Since these recordings reflect realistic environments rather than controlled data collection, they naturally contain a higher number of private vehicles.

Frame extraction at 1 FPS captured scenes as they occurred without selectively increasing or decreasing specific object categories. This ensured realism but also preserved the natural imbalance present in traffic conditions.

Additionally, the confidence threshold was set to 85% during the auto-labeling process. This means that only detections with high confidence were accepted automatically. While this setting helps reduce incorrect annotations, it may slightly reduce the number of detected objects in smaller or less visible classes, such as buses or motorcycles. However, the major cause of imbalance remains the natural frequency of vehicles in real-world traffic scenes rather than the threshold setting itself. Therefore, the imbalance observed in the dataset is not due to annotation errors but reflects authentic traffic patterns.

IV. Effects of Data Imbalance

Data imbalance can significantly influence how a machine learning model learns from the dataset. When one class appears much more frequently than others, the model

may prioritize learning patterns from the dominant class while paying less attention to minority classes.

In this dataset created, the Car class contains 6,187 instances, which is substantially higher than other classes such as Bus (171 instances) and Motorcycle (277 instances). Because of this imbalance, the model may become very accurate in detecting cars but less accurate when identifying buses, motorcycles, or trucks.

This can lead to the following:

- Lower detection accuracy for underrepresented classes
- Higher false negatives for rare objects
- Bias toward predicting the dominant class
- Reduced fairness in model performance across all categories

Although the imbalance reflects realistic traffic conditions, it may still affect the overall performance of the object detection model if not properly managed during training.

V. Methods Used to Modify and Manage Imbalance

Although the dataset reflects a natural traffic imbalance, several steps were taken to manage its potential impact. First, frames were extracted at a controlled rate of 1 frame per second to reduce repeated scenes where dominant objects such as cars appear continuously. This helped minimize unnecessary duplication and improve variation between frames. During the review stage of the annotation, similar or low-quality frames were removed to prevent excessive repetition of dominant-class objects. In addition, controlled data augmentation techniques such as exposure adjustment and noise addition were applied to increase dataset diversity and improve model generalization. While augmentation was not specifically targeted at minority classes, it helps reduce overfitting and encourages the model to learn from a wider range of visual conditions. These steps helped manage imbalance while still preserving the realistic distribution of objects in traffic scenes.

VI. Algorithm Approaches

In addition to dataset-level adjustments, algorithm-based techniques can be applied during model training to further reduce the impact of class imbalance. One common approach is class weighting, where higher importance is assigned to underrepresented classes during loss calculation. This encourages the model to pay more attention to minority classes such as buses and motorcycles. Another method is focal loss, which reduces the influence of easily predicted dominant classes and focuses more on harder or less frequent objects. Weighted sampling can also be used during training, where images containing minority classes are selected more frequently in training batches. These algorithm-level strategies help reduce bias toward dominant classes without altering the original dataset distribution and can improve detection performance across all object categories.

Reference

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